UML Diagram Suite

Resource-Aware Collaborative Cloud IDE

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Department of Computer Engineering
Academic Year 2026–2027

1. Use Case Diagram
2. Class Diagrams (Collaboration, Provisioning, Optimization)
3. Sequence Diagram (two parts)
4. State Machine Diagram
5. Activity Diagram
6. Component Diagram
7. Deployment Diagram
8. Object Diagram
9. Collaboration (Communication) Diagram
10. Package Diagram
11. Composite Structure Diagram

1. Use Case Diagram

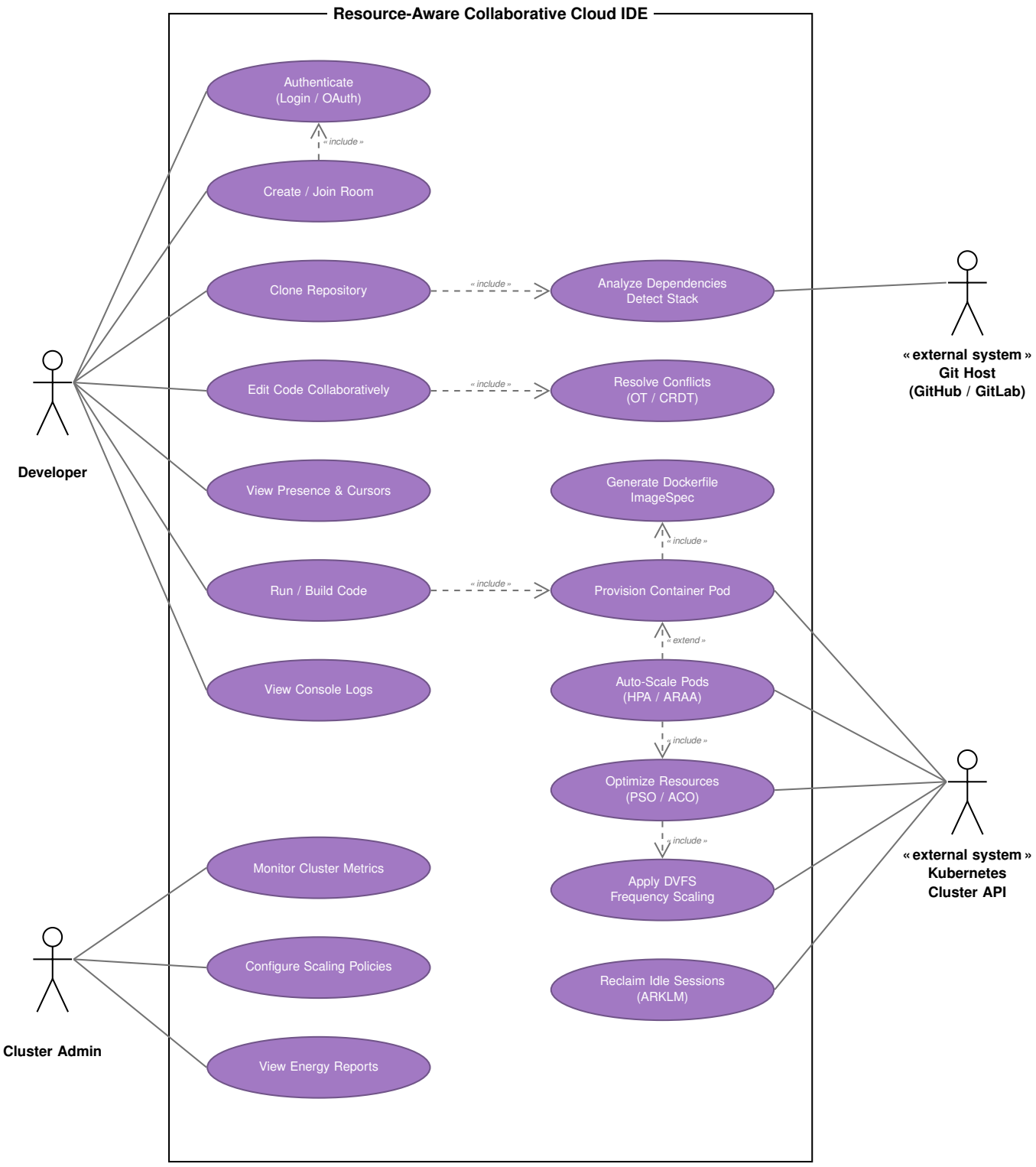


Figure 1: Use Case Diagram

2A. Class Diagram: Collaborative Real-Time Workspace

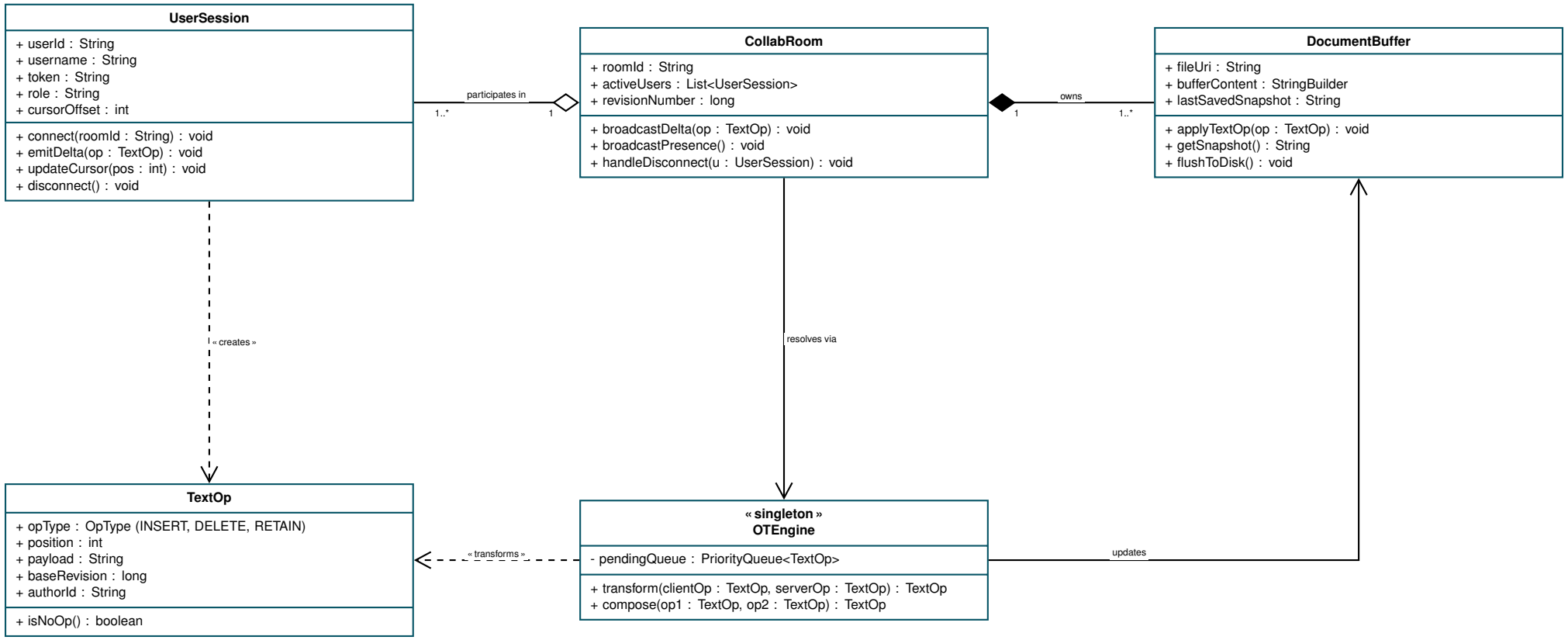


Figure 2: Class Diagram: Real-Time Collaborative Engine

2B. Class Diagram: Workspace Provisioning and Runtime Execution

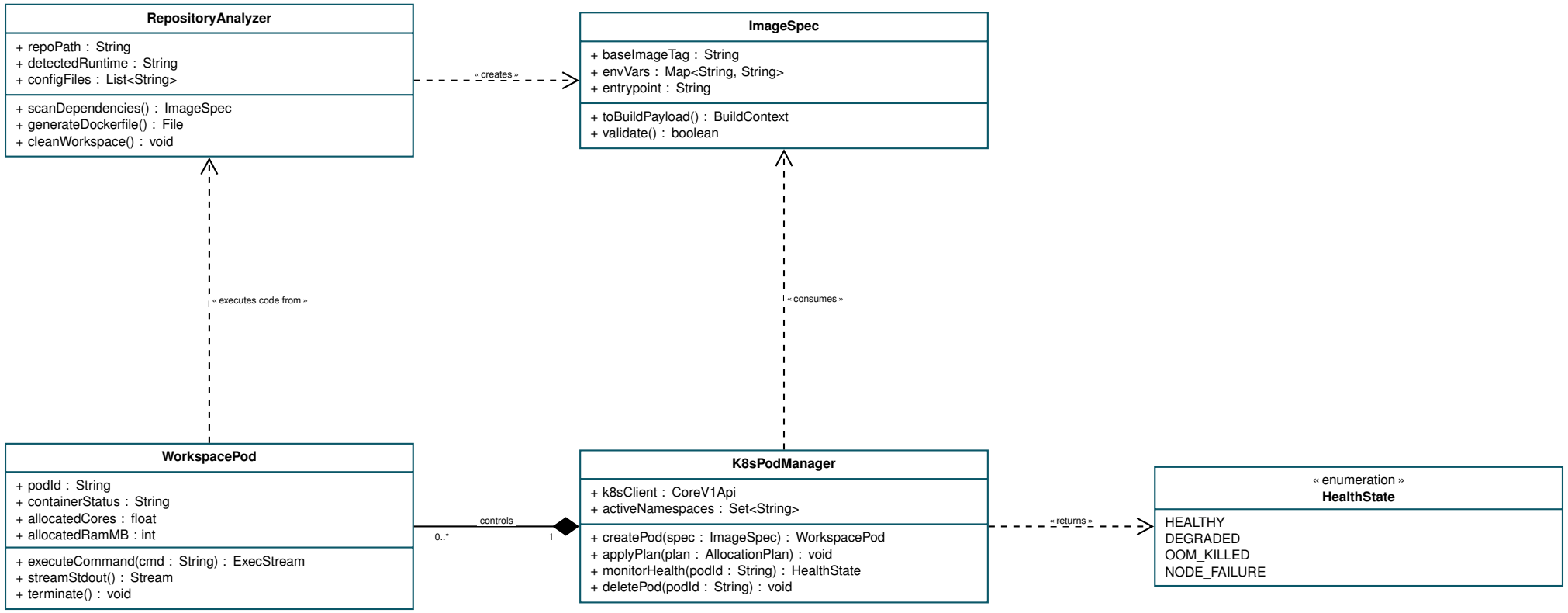


Figure 3: Class Diagram: Workspace Container Provisioning & Runtime

2C. Class Diagram: Heuristic Optimization and Lifecycle

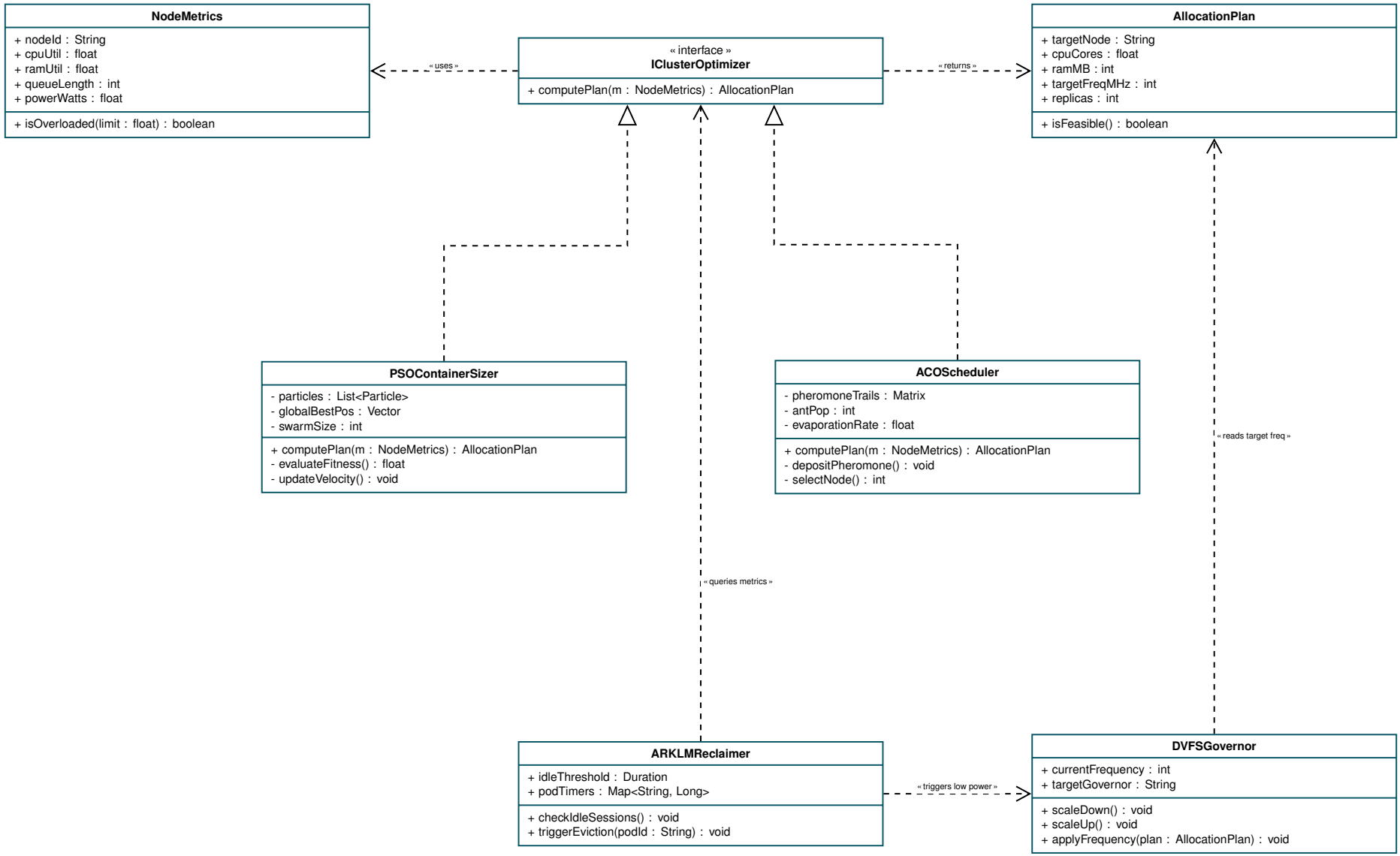


Figure 4: Class Diagram: Heuristic Optimization & Lifecycle Engine

3A. Sequence Diagram: Session Setup and Real-Time Collaboration

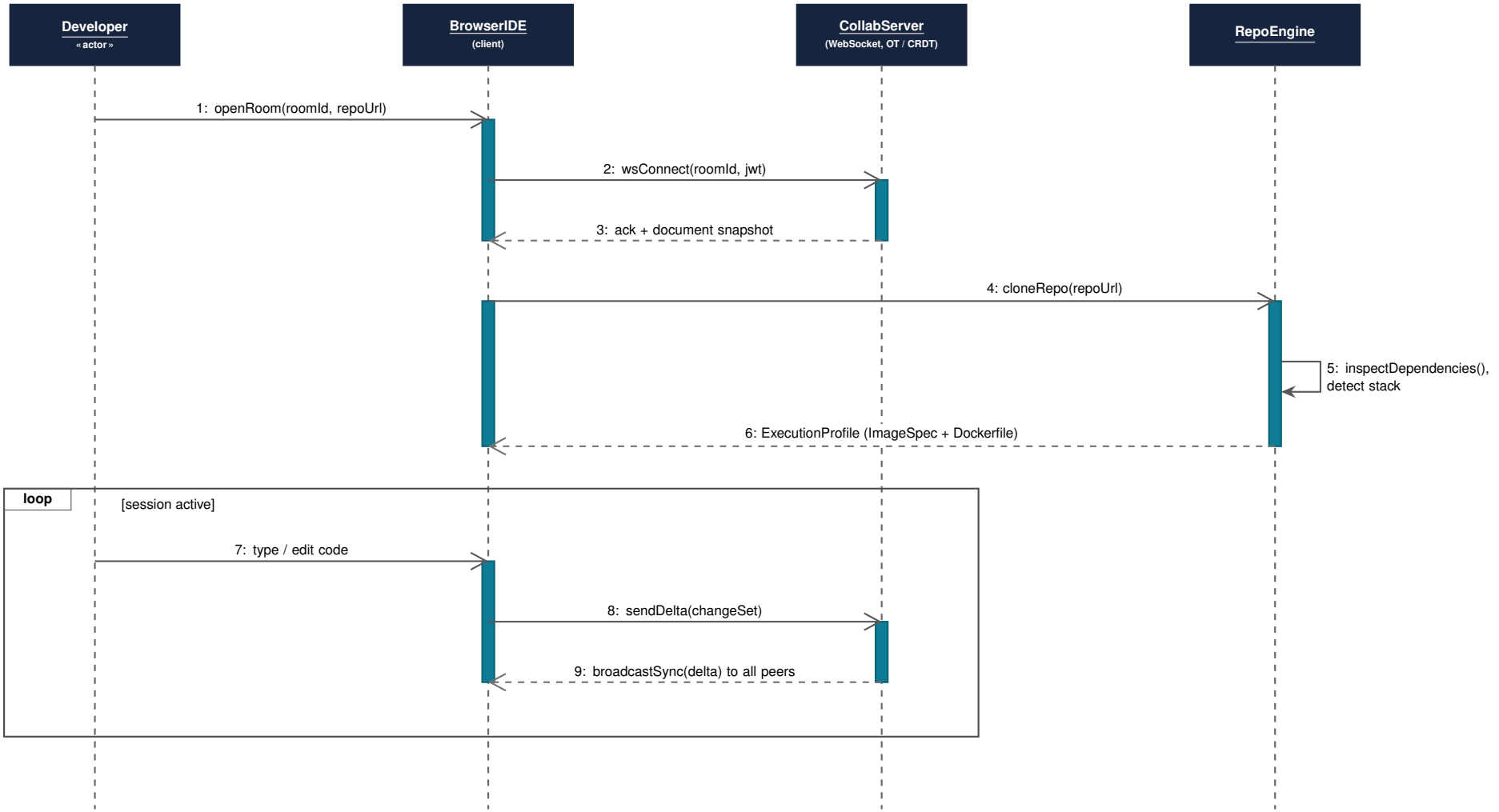


Figure 5: Sequence Diagram, part A: room join, repository analysis and collaborative editing (messages 1–9)

3B. Sequence Diagram: Execution, Scaling and Teardown

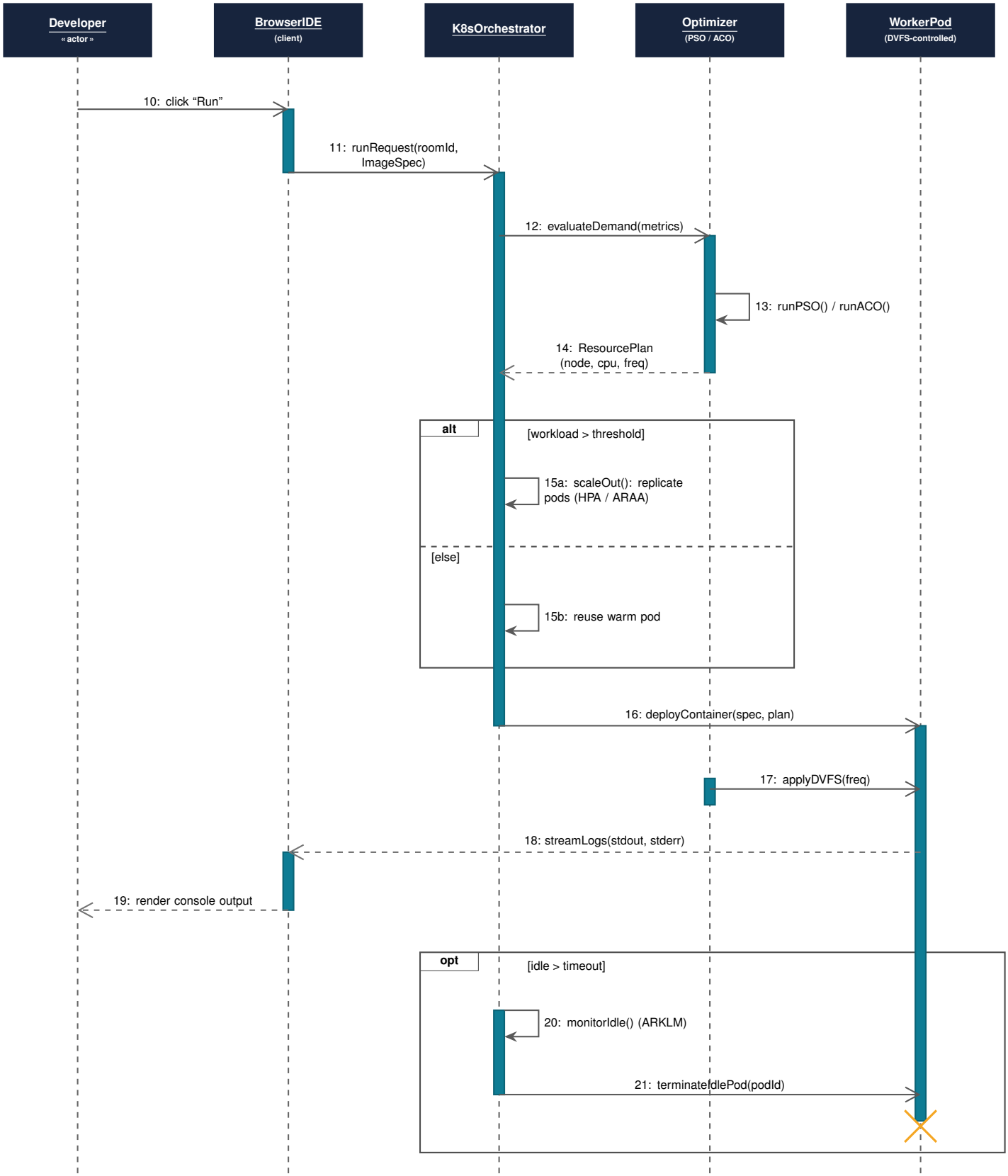


Figure 6: Sequence Diagram, part B: run request, optimization, scaling, log streaming and idle reclaim (messages 10–21)

4. State Machine Diagram: Workspace Container Lifecycle

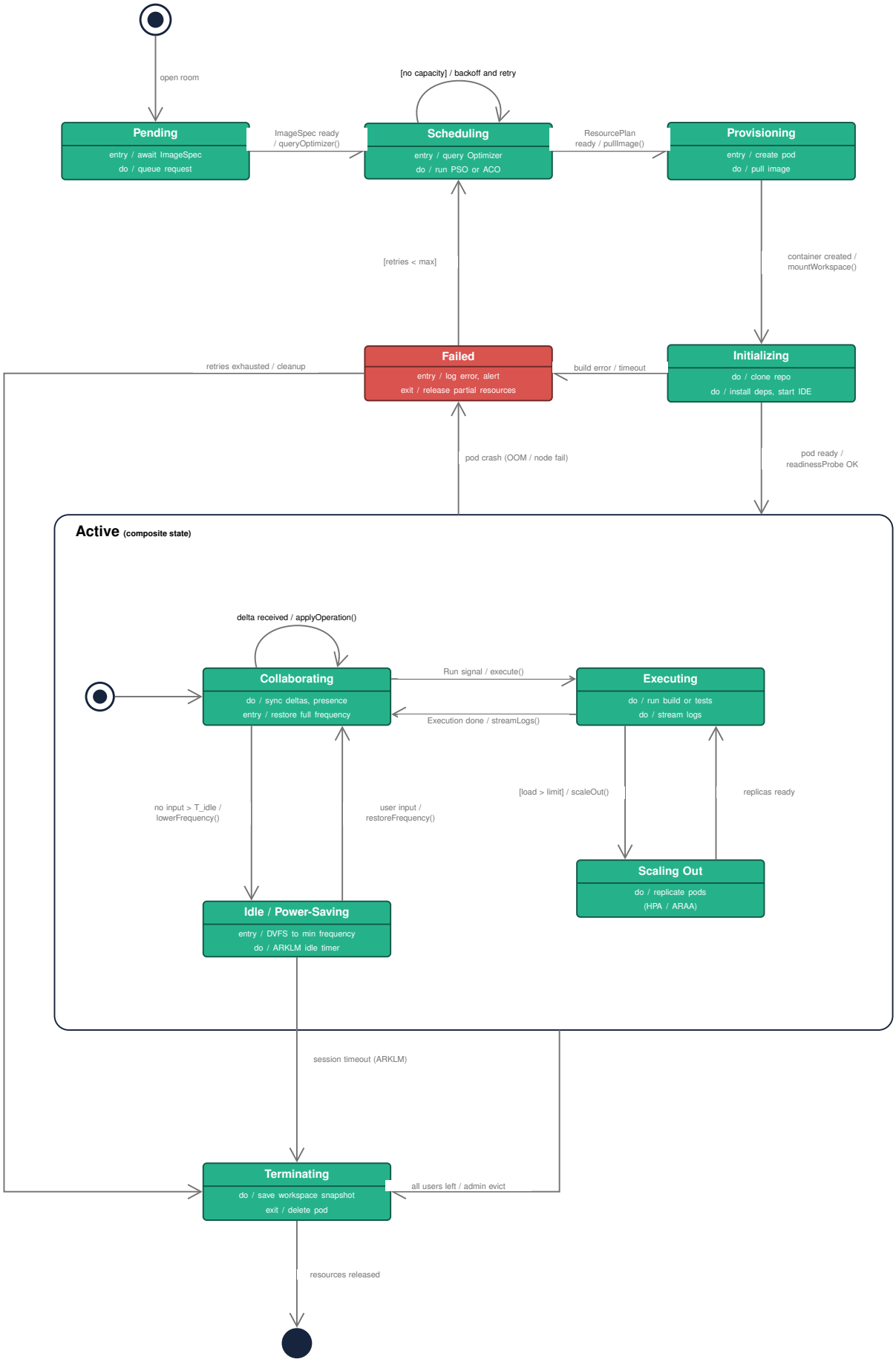


Figure 7: State Machine Diagram: Workspace Container Lifecycle

5. Activity Diagram: Resource Allocation and Execution

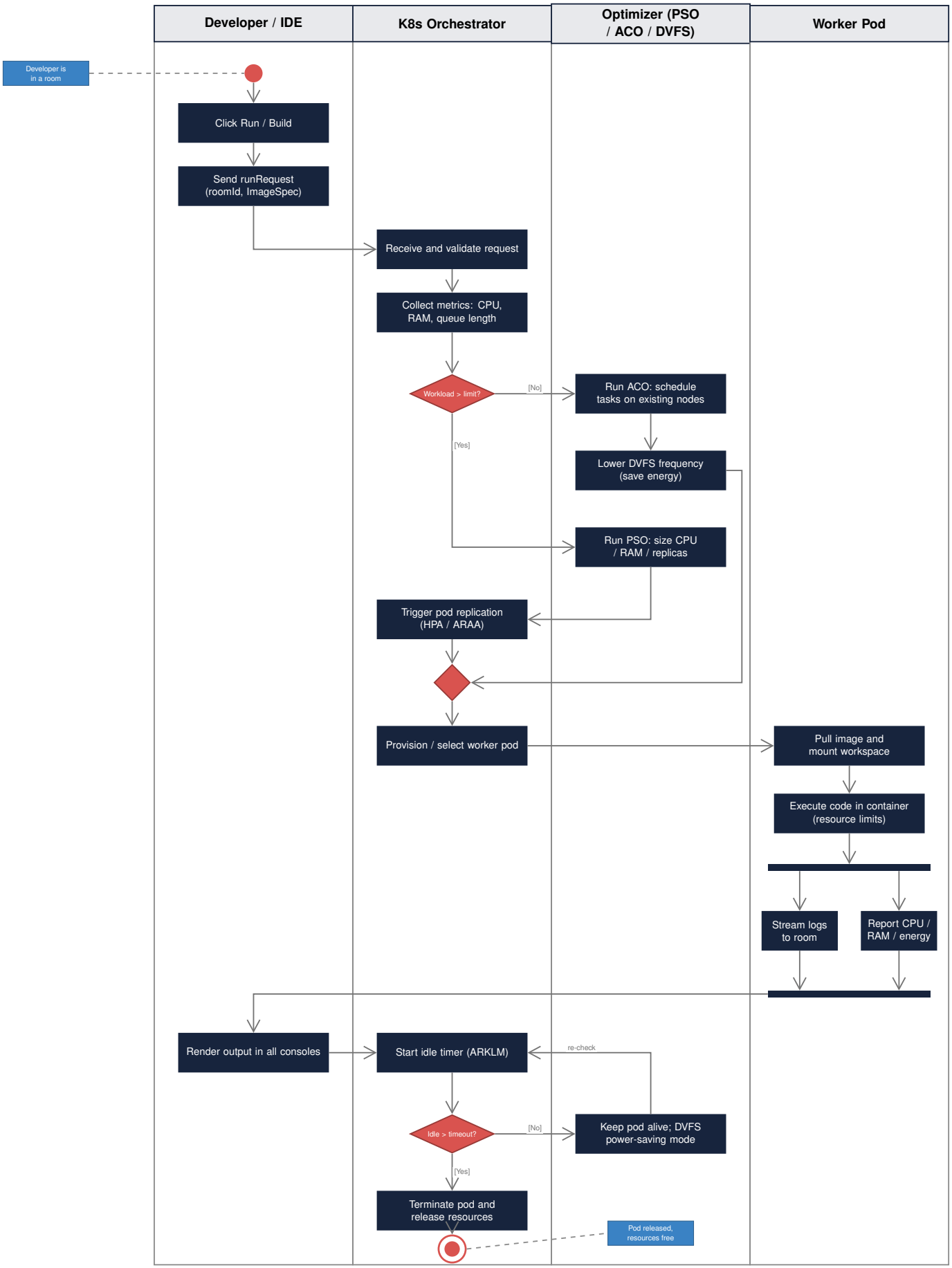


Figure 8: Activity Diagram: Resource Allocation and Execution

6. Component Diagram

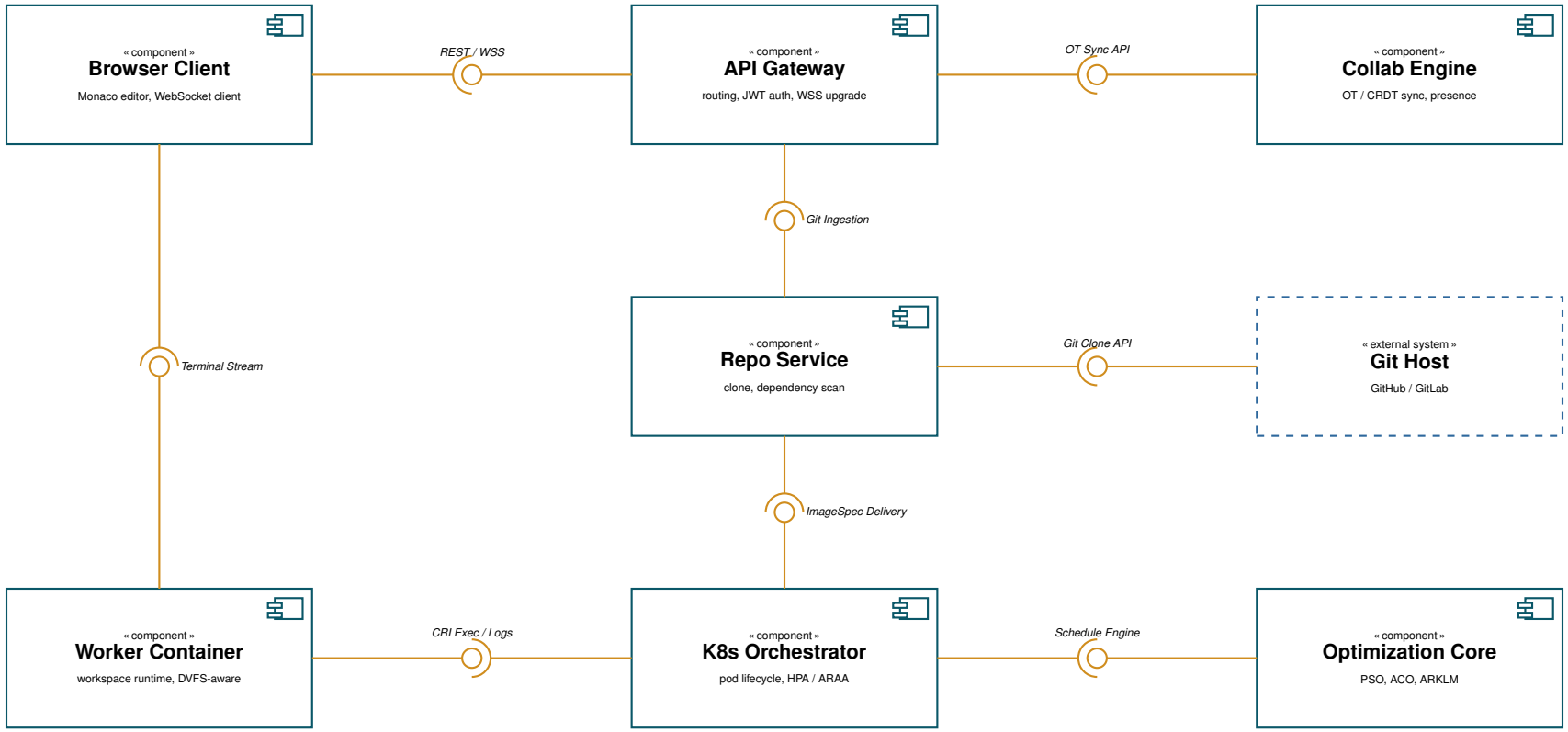


Figure 9: Component Diagram (ball = provided interface, socket = required interface)

7. Deployment Diagram

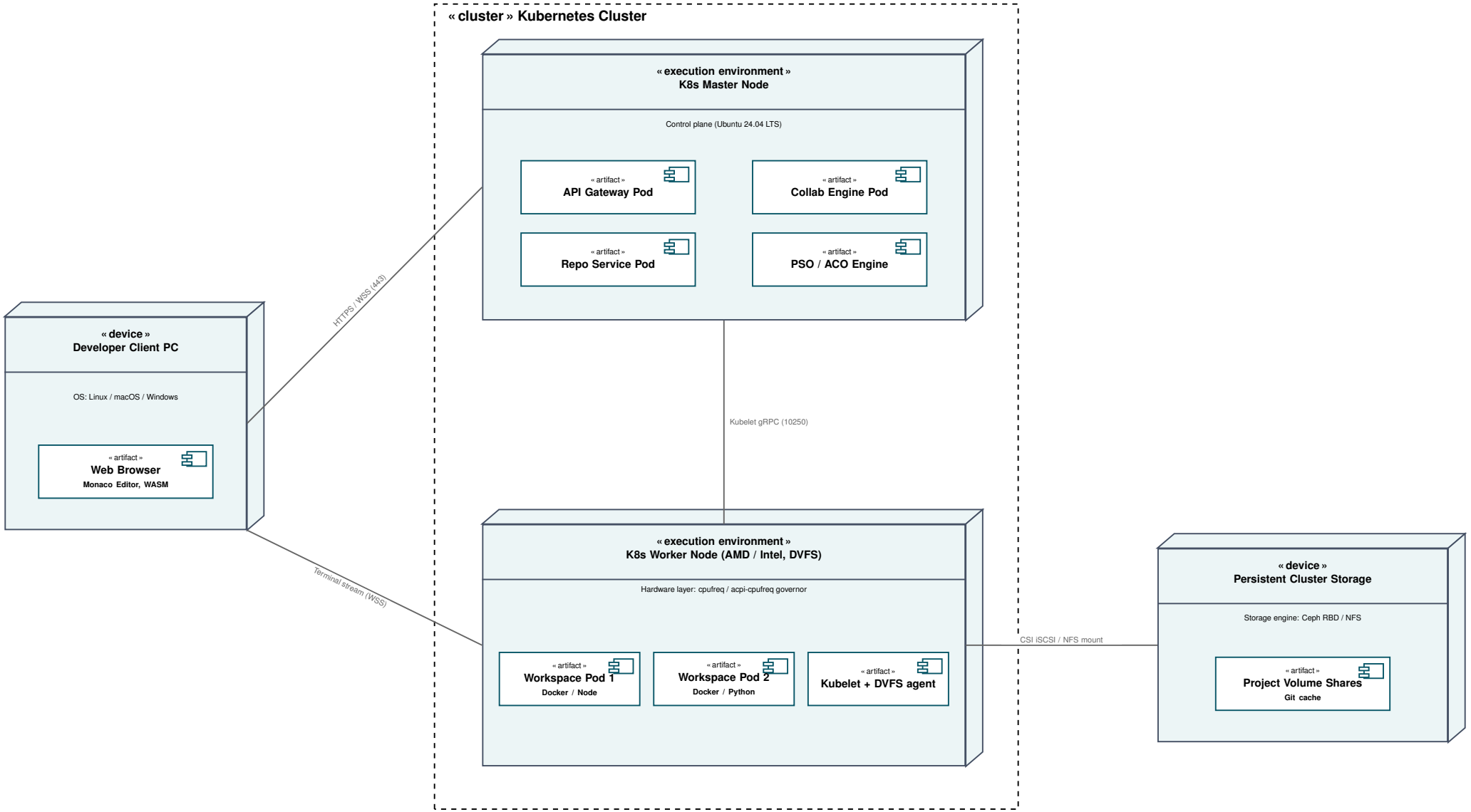


Figure 10: Deployment Diagram

8. Object Diagram (Runtime Instance Snapshot)

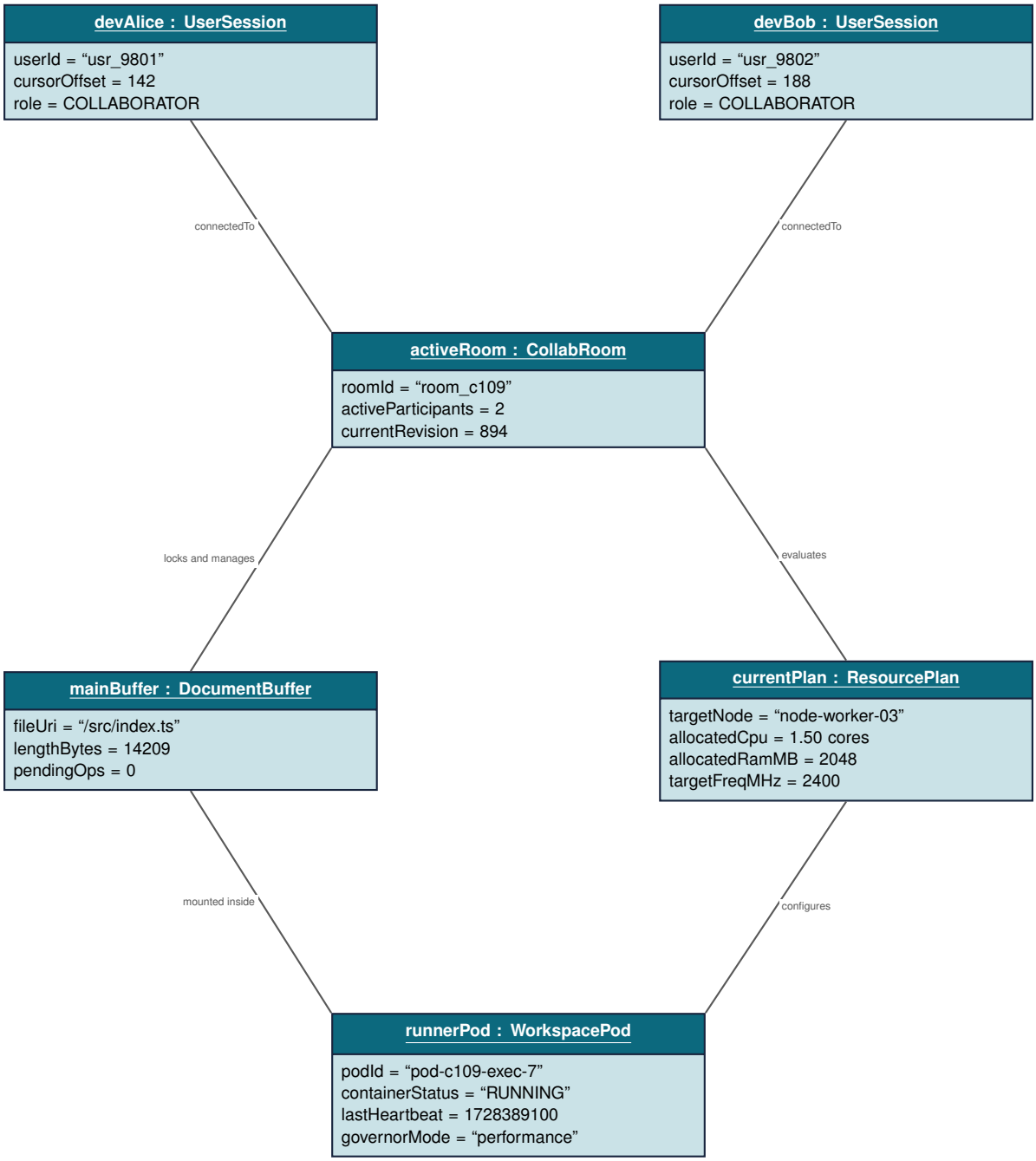


Figure 11: Object Diagram (Runtime Instance Snapshot)

9. Collaboration (Communication) Diagram

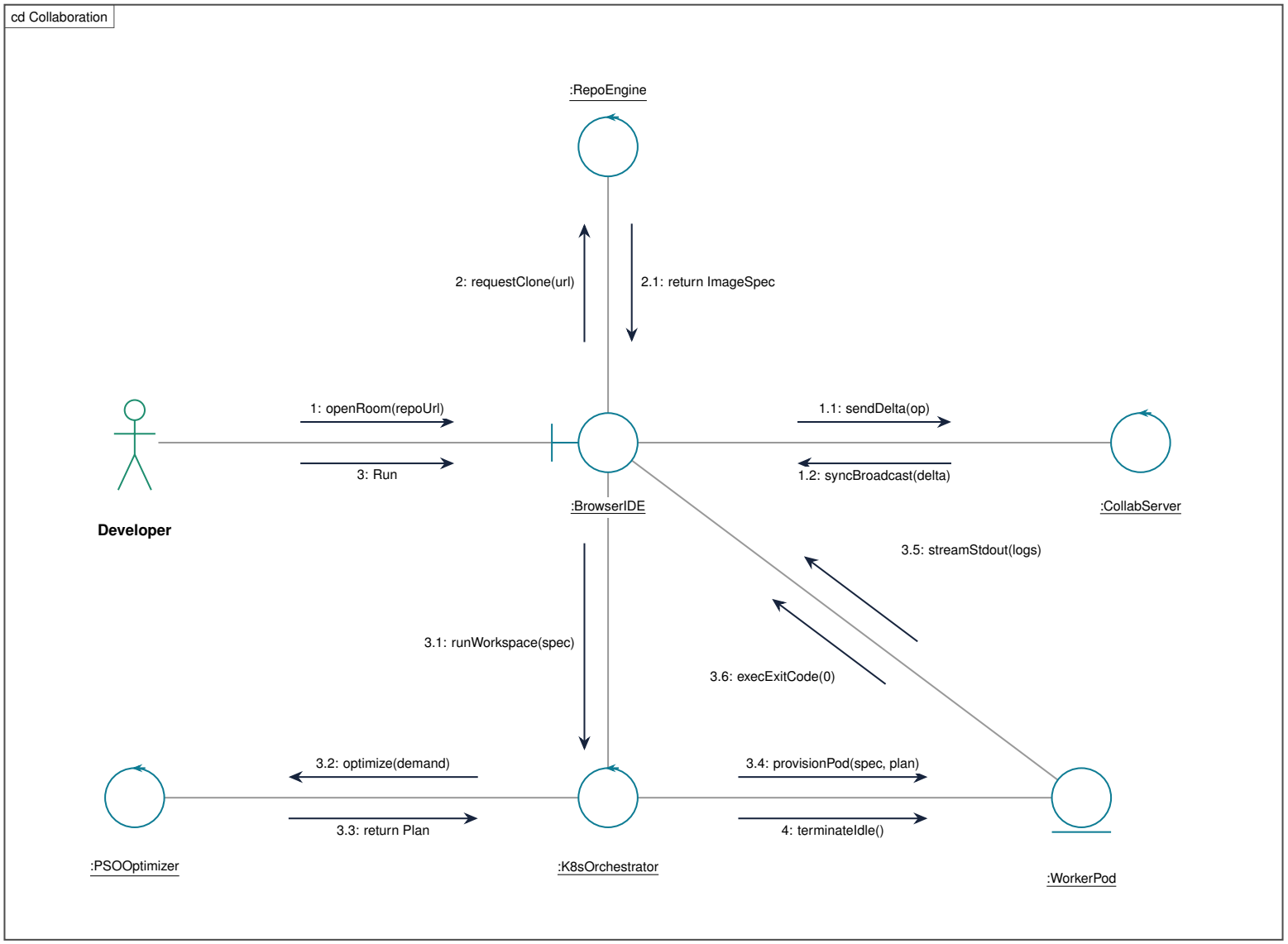


Figure 12: Collaboration (Communication) Diagram

10. Package Diagram

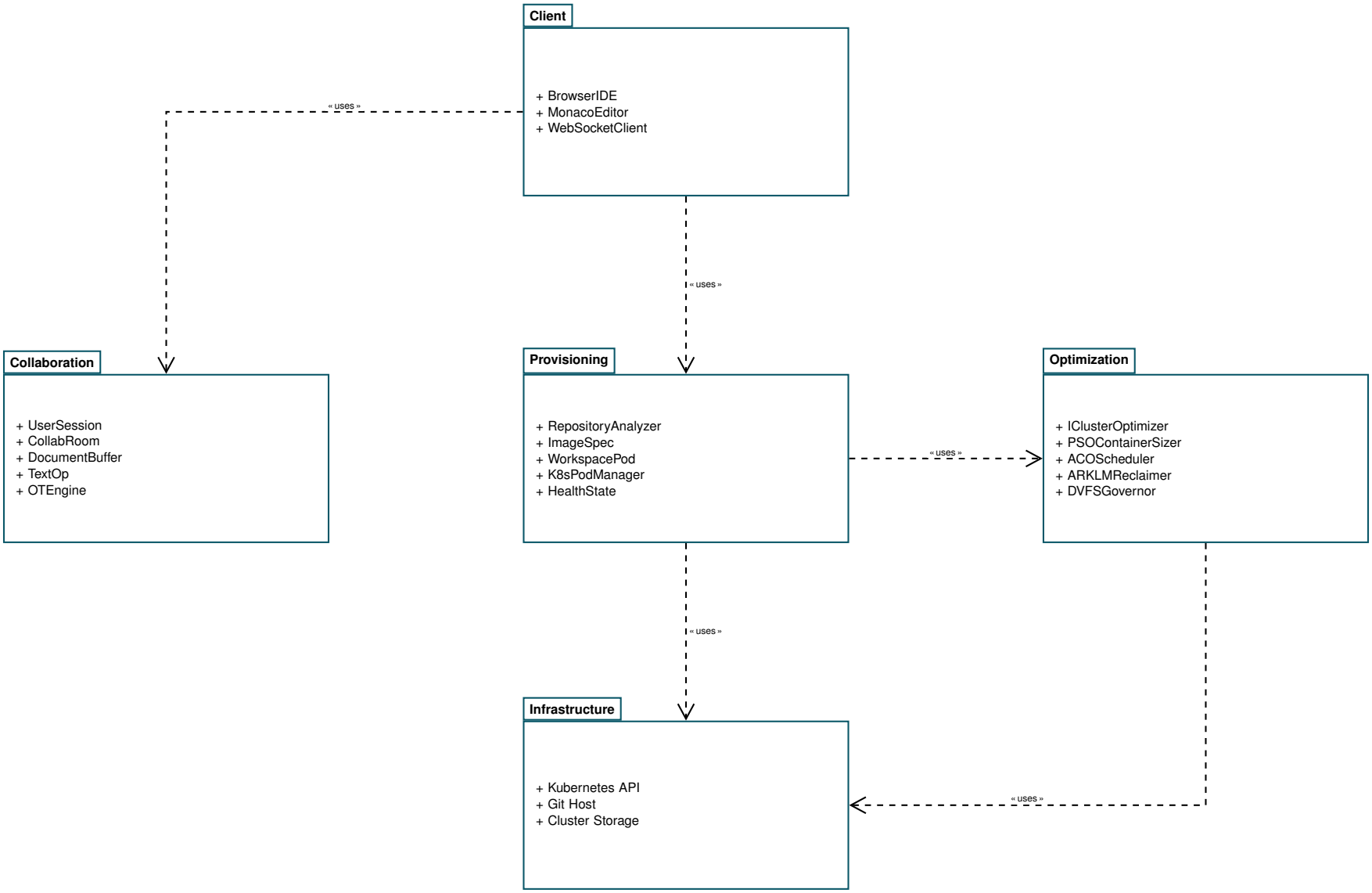


Figure 13: Package Diagram: dependencies between the platform's packages

11. Composite Structure Diagram

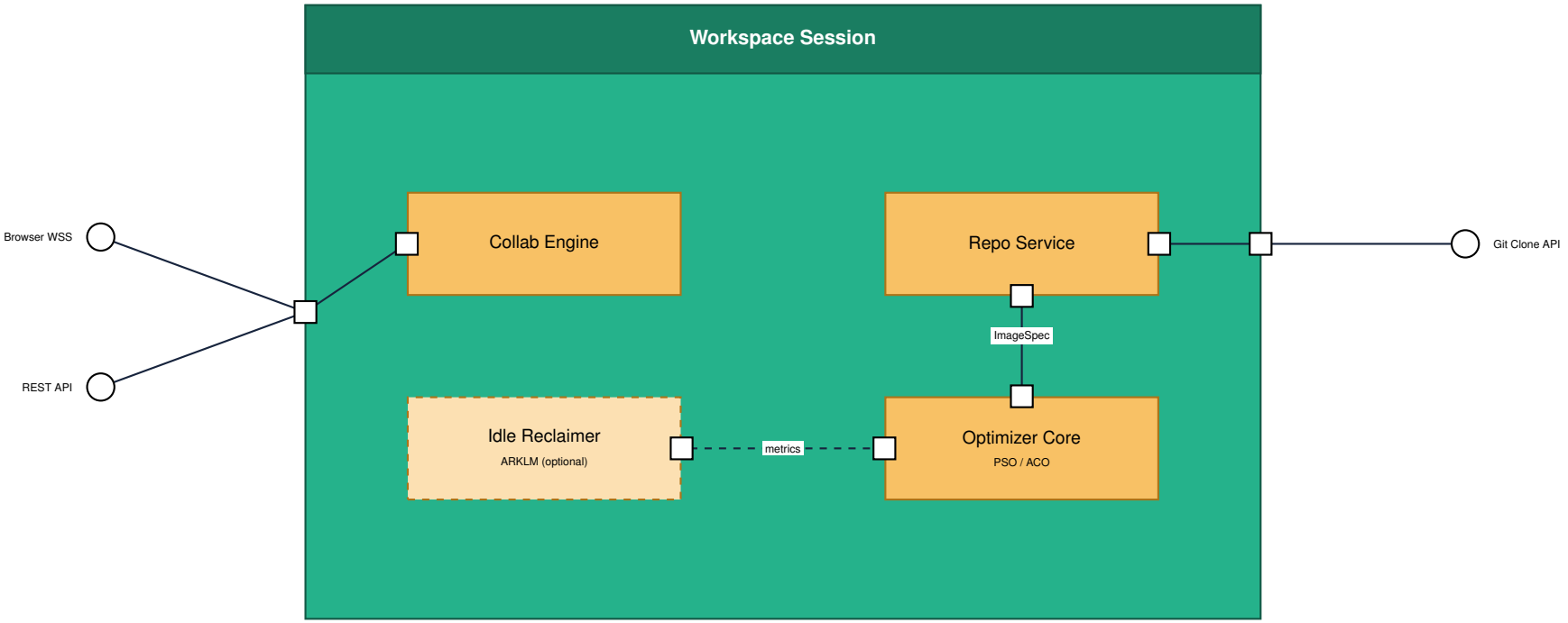


Figure 14: Composite Structure Diagram: internal structure of a workspace session